

SambaNova SN50 Hot Chips 2026 SN50 Dataflow

For as new as the dedicated AI accelerator field is, SambaNova is one of the older and more established hardware vendors. The company is now in the fifth generation of their reconfigurable dataflow unit (RDU) technology, with the SN50 that was launched earlier this year. As with the other major AI vendors at this year’s Hot Chips conference, the company has come to present new technical details on SN50, and outline what makes it competitive in the burgeoning field of dedicated AI accelerators.

SambaNova’s hardware has taken on an increased prominence in the industry thanks to the company’s connection to Intel. While Intel itself is still trying to catch up on AI accelerators, the company has become increasingly attached to the hip to SambaNova, whose RDUs provide the dedicated, high-efficiency and low-latency AI accelerators that round out Intel’s hardware stack. Thus the company’s progress with the SN50 (and future RDUs) is material not just for SambaNova, but for Intel as well.

Agentic inference is decode-dominated

All AgentX frontier models; DeepSeek-V3 8K/1K reference included



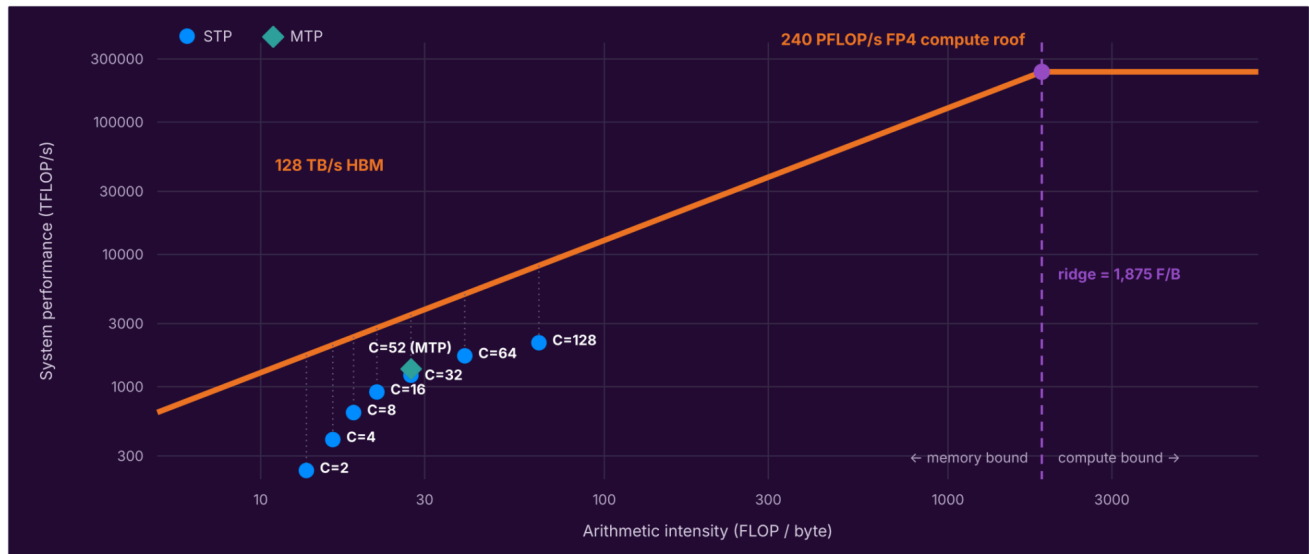
AgentX, Cao4, et. al.; decode = (OSL-1) * TPOT. DeepSeek-V3 reference: 8K input / 1K output

SambaNova SN50 Hot Chips 2026 Inference Decode

Setting the stage, agentic inference is all the rage right now. Where does all the execution time go? SambaNova has a breakdown of it. Most time is spent in decode, especially on DeepSeek V3 where it’s 97% of the time, versus 3% for prefill.

Decode is Bandwidth-Bound

DeepSeek-R1 8K/1K • B300 FP4 • InferenceX measured data • STP + MTP3 • 16 decode GPUs



SambaNova SN50 Hot Chips 2026 Decode Bandwidth Bound

And decode, in turn, is bandwidth-bound. The FLOPS-per-byte ratio is quite low, even for large batch sizes.

HBM Bandwidth Utilization ≠ Model Bandwidth Utilization (MBU)

MBU = fraction of peak HBM bandwidth used for weights and KV caches

$$\text{tokens/s} = \frac{\text{HBM bandwidth/chip} \times \text{MBU} \times \text{number of chips}}{\text{parameter bytes} + \text{KV-cache bytes/token}}$$

Peak HBM bandwidth

What the hardware can move

MBU

What the model actually uses of it

Token speed

Follows MBU, not just chip count

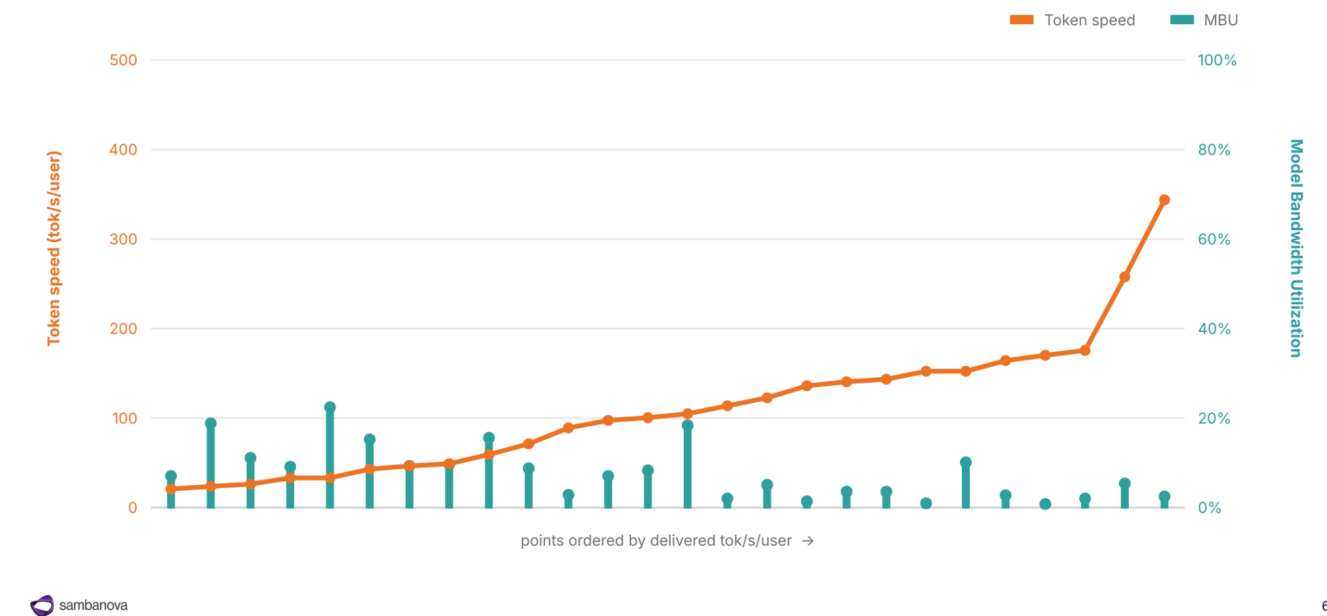
SambaNova SN50 Hot Chips 2026 HBM Bandwidth Utilization

Bandwidth utilization is often misunderstood. SambaNova is laying out what they mean for this talk. In short, they aren't talking about just how much HBM bandwidth is being used, but rather

the Model Bandwidth Utilization (MBU) model. And specifically, what fraction of that is being used to cache data and otherwise handle data usage.

GPUs: Low Model Bandwidth Utilization

DeepSeek-R1 8K/1K | B300 FP4 | all 26 InferenceX points | sorted by delivered tok/s/user

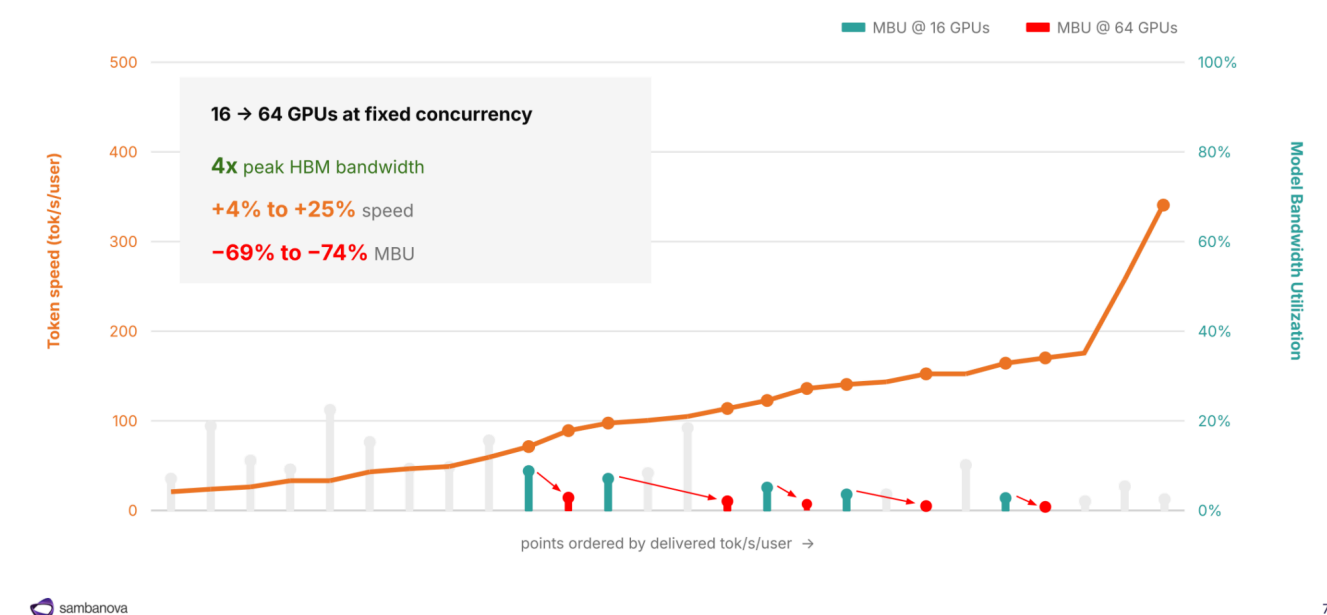


SambaNova SN50 Hot Chips 2026 Model Bandwidth Utilization

Looking at the current state of tech, GPUs offer low bandwidth usage, even with highly optimized GPU-friendly benchmarks.

Increasing GPUs from 16 → 64: Higher Speed, Lower MBU

DeepSeek-R1 8K/1K | B300 FP4 | 26 InferenceX GPU points | sorted by delivered tok/s/user

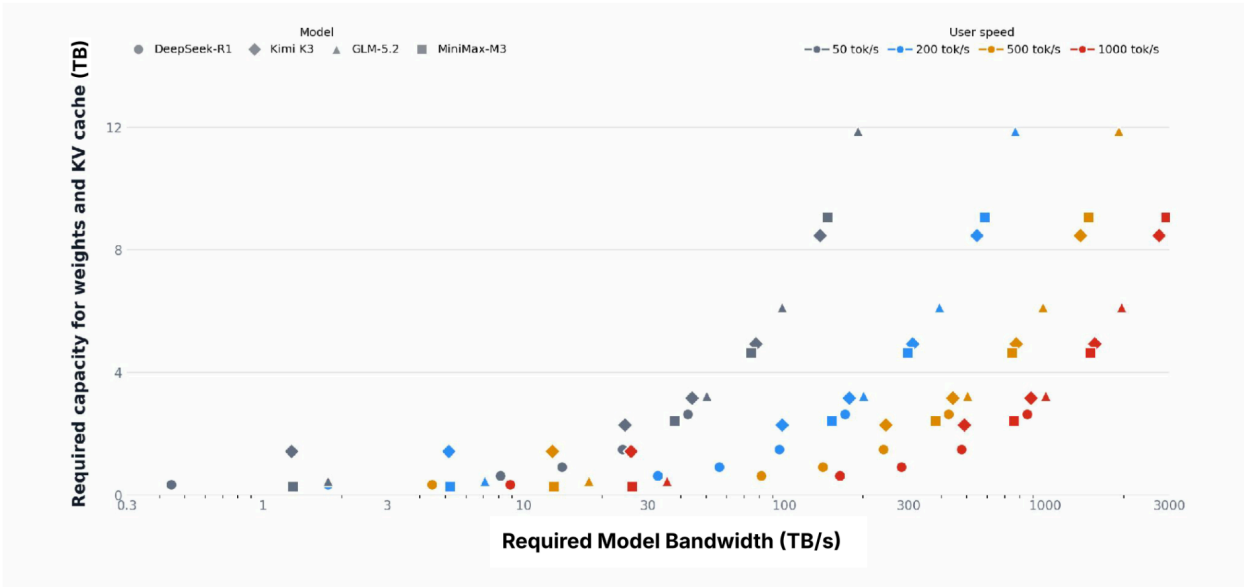


SambaNova SN50 Hot Chips 2026 GPU Scaling

Things get worse for GPUs when you scale up the number of them; performance does go up, but MBU drops significantly.

Capacity and MBU: Frontier Models Demand Scale

DeepSeek-R1, Kimi K3, GLM-5.2, MiniMax-M3 | points only | MTP3 accepted length = 3.1

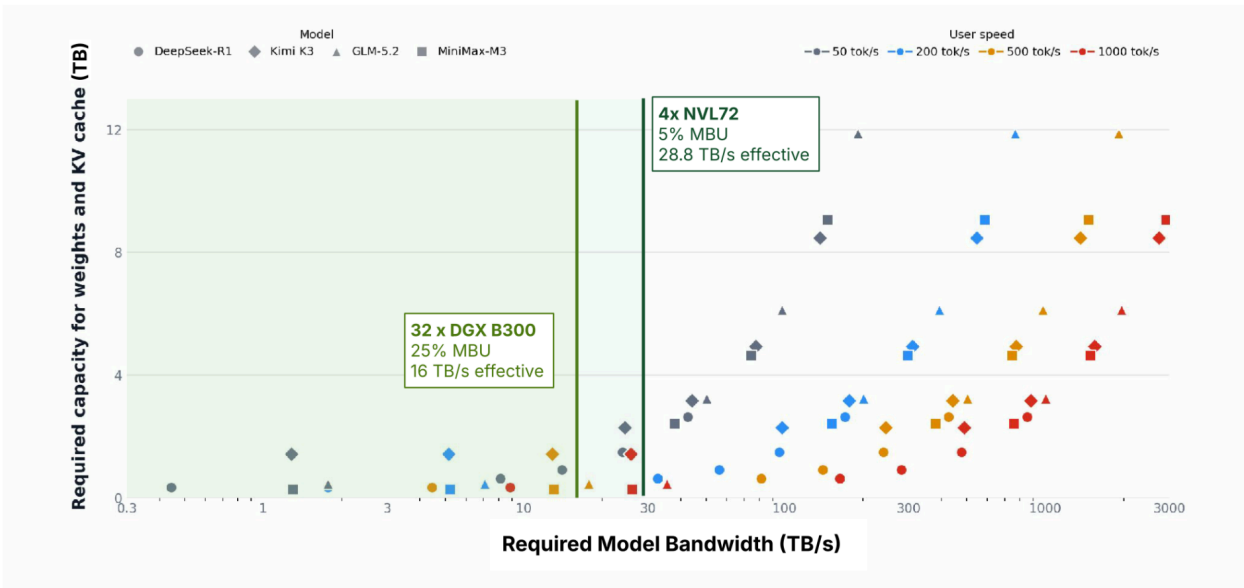


SambaNova SN50 Hot Chips 2026 Frontier Models

Meanwhile frontier models require being able to scale up.

GPUs @500 kW: Power Buys Capacity, not Bandwidth

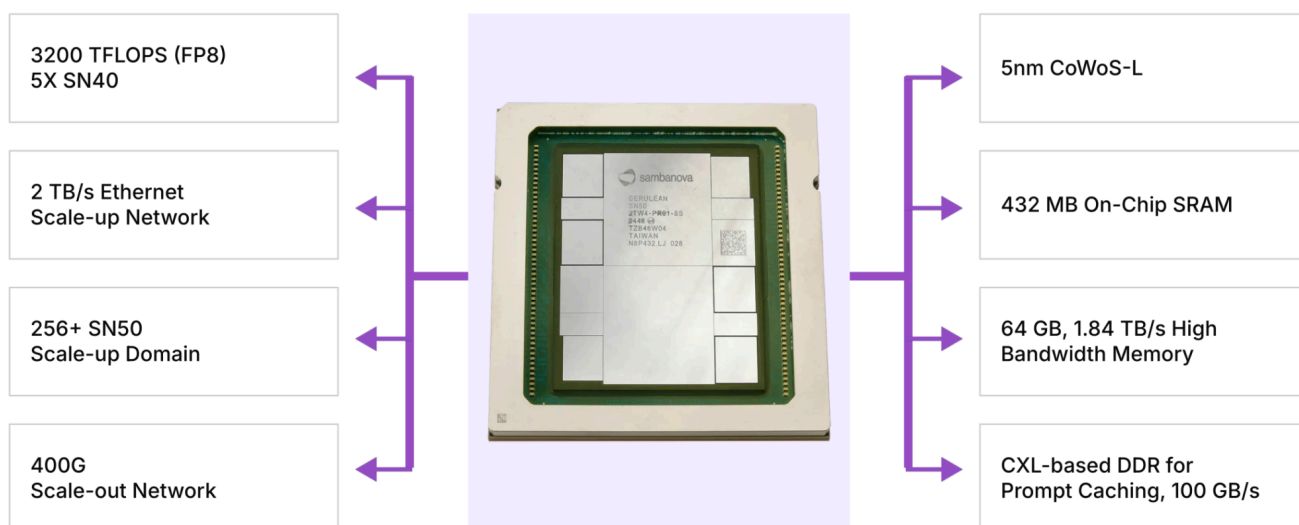
All frontier-model points | MTP3 accepted length = 3.1



SambaNova SN50 Hot Chips 2026 Power Capacity

Again with a GPU example, a GPU can get to around 30TB/second of model bandwidth. But they can't get past that.

SN50: DataFlow Built For Scale



SambaNova SN50 Hot Chips 2026 SN50 Dataflow

Enter SambaNova's SN50 dataflow RBU. They have doubled-down on what worked well from SN40, such as the large on-chip SRAM. 5x as many FLOPS as SN40, and it is designed to scale-up to a much larger domain of 256+ chips. And there is a separate scale-out network using 400Gb networking.

Notably, there are no I/O dies or similar here. Instead it is just two max reticle dies for the logic, and then HBM stacks for the memory.

Though it is interesting that SambaNova's choice of HBM here is quite dated; SN50 still uses HBM2e here (which is going to be a problem in the future as production of the memory is already ramping down).

SN50 Air-Cooled Rack Architecture

Total per Rack: 16 RDUs, 2 Nodes

- Compute: 25.6 PFLOPs @BF16, 51.2 PFLOPs @FP8
- SRAM: 6.9 GB
- HBM: 1 TB
- RDU DDR: 256 GB to 2 TB
- Host: 2× 64-core CPUs, 1 TB DDR5
- Host boot and storage: 4× 960 GB NVMe SSD
- Host storage: 6× 7.6 TB NVMe SSD
- Networking
 - Scale-up: 6.4 TB/s
 - Scale-out: 800 GB/s
 - Host: 4×100G
- Management: 1 GbE switch / serial console server
- Power Consumption:
 - Typical: 15 - 20 kW
 - Maximum: Up to 34 kW
- Cooling: Air-cooled
- Rack Dimensions:
 - Height 2026 mm, Width 600mm, Depth 1272mm (including doors)

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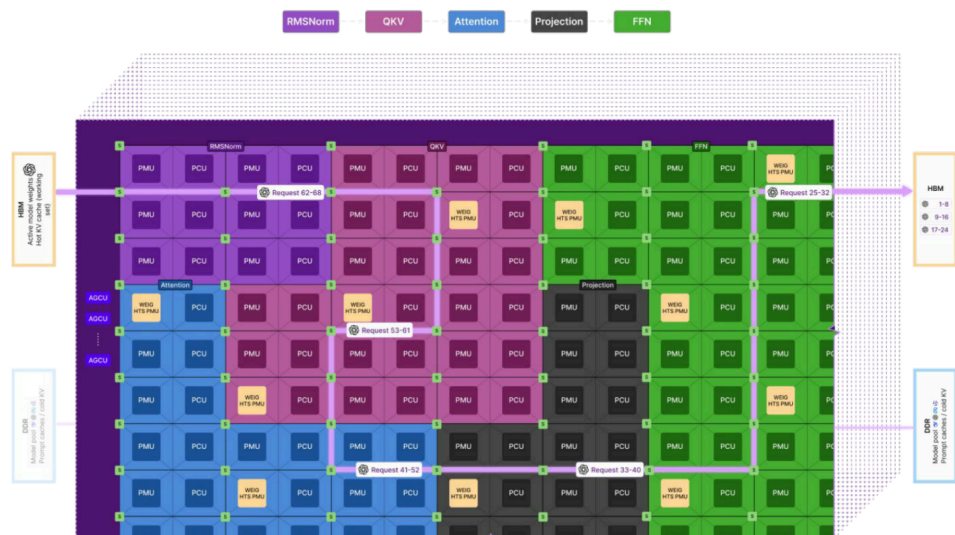
SambaNova SN50 Hot Chips 2026 SN50 Rack

Moving up to the SN50 rack architecture, there are 16 RDUs in a single air-cooled rack, split over two nodes.

SN50: Chip Architecture

864 PCUs and PMUs

- PCU**: Compute unit
- PMU**: Memory unit
- S**: Mesh switches
- AGCU**: Portal to off-chip memory and I/O

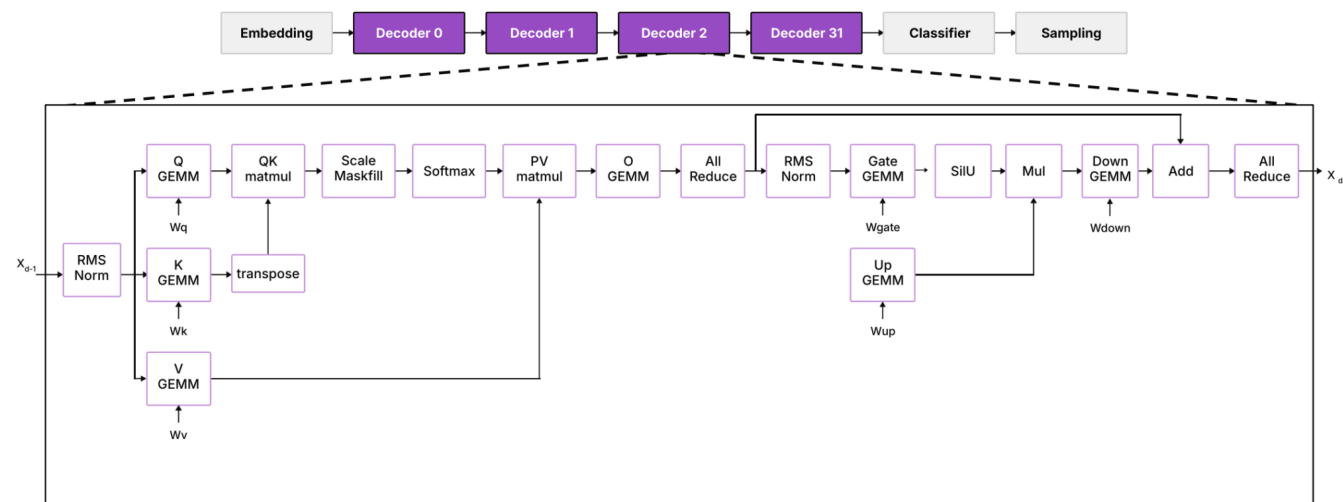


SambaNova SN50 Hot Chips 2026 Chip Architecture

Diving a bit deeper into SN50 and the dataflow architecture. The core element of the SN50 is the sea of compute cores (PCUs) and memory cores (PMUs). There is no hardware memory

management; this is all software managed. Every unit operates when it has input and sends it to the outputs.

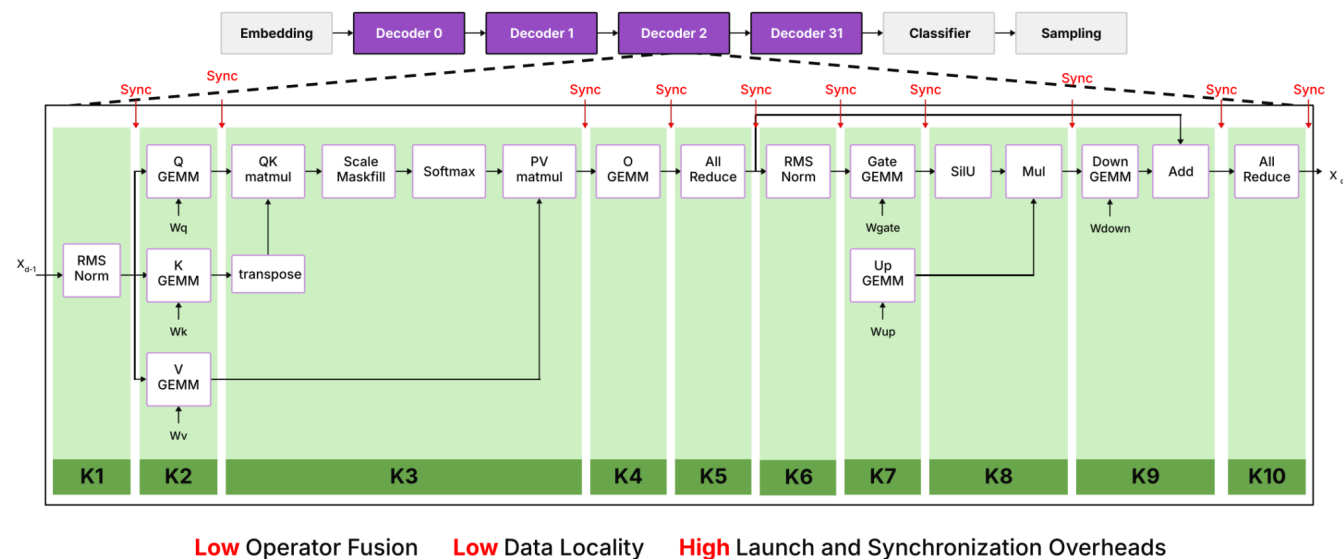
Structure of a Transformer



SambaNova SN50 Hot Chips 2026 Transformer Structure

To better illustrate how the dataflow architecture works, here is an example of how it maps to a transformer.

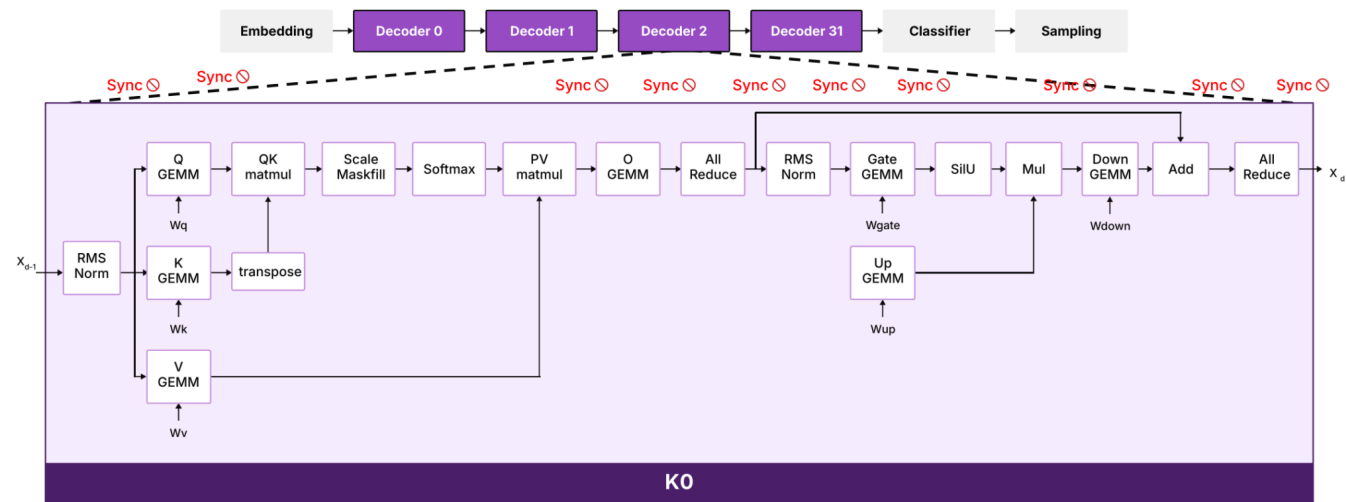
GPUs Incur Overheads with Lower Operator Fusion



SambaNova SN50 Hot Chips 2026 GPU Transformers

Here is what a GPU looks like.

SN50: Persistent Decoder Kernel, No Overheads!

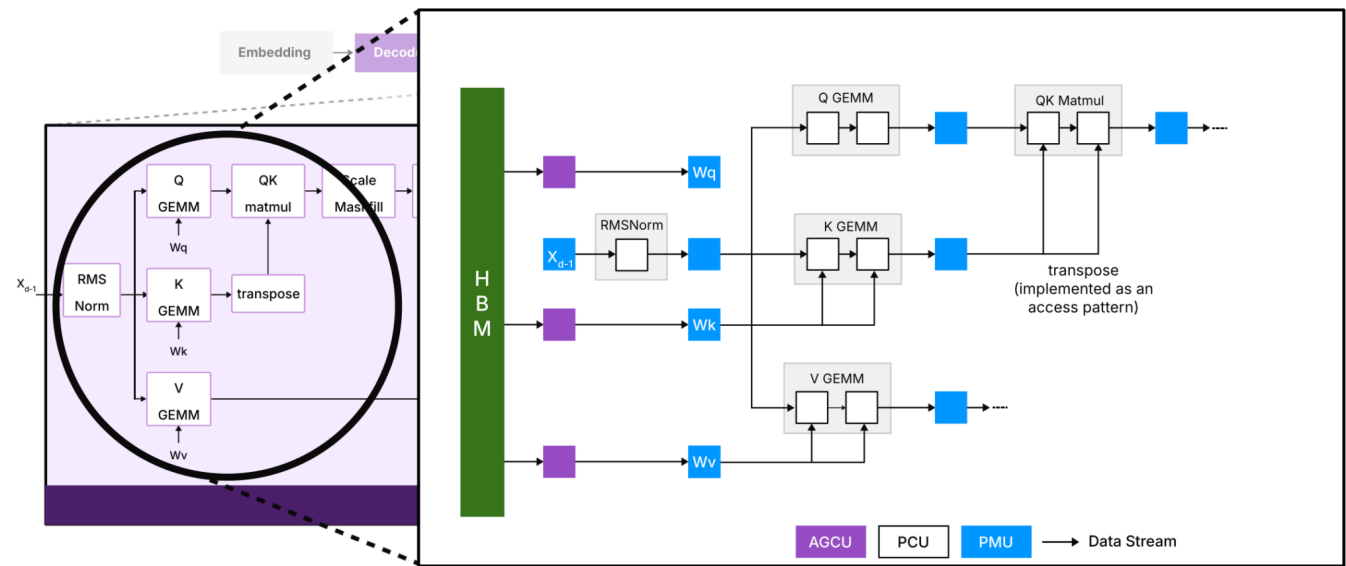


High Operator Fusion: One Kernel Call for **All Decoders!** **Zero** Kernel Launch Overheads **High** Data Locality

SambaNova SN50 Hot Chips 2026 SN50 Transformers

And how it look on the SN50.

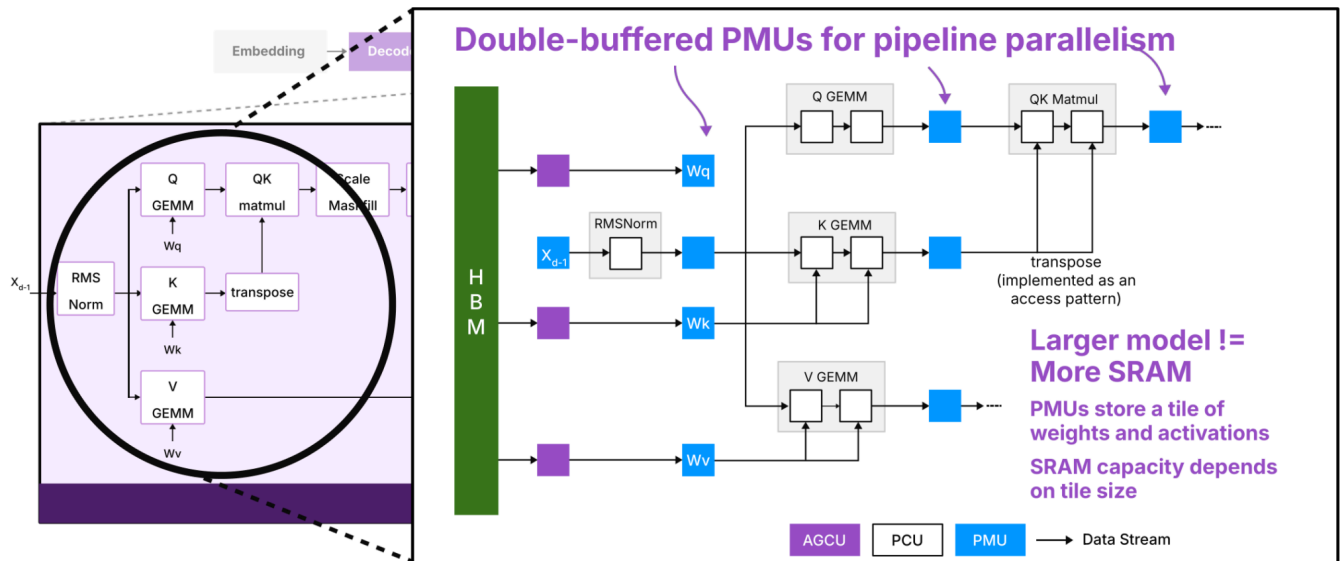
Compute - Communication Overlap: Within RDU



SambaNova SN50 Hot Chips 2026 Compute Comms Overlap

For compute, data from the HBM is fed into the AGCU portal that control off-chip access, and from there into the PCUs and PMUs.

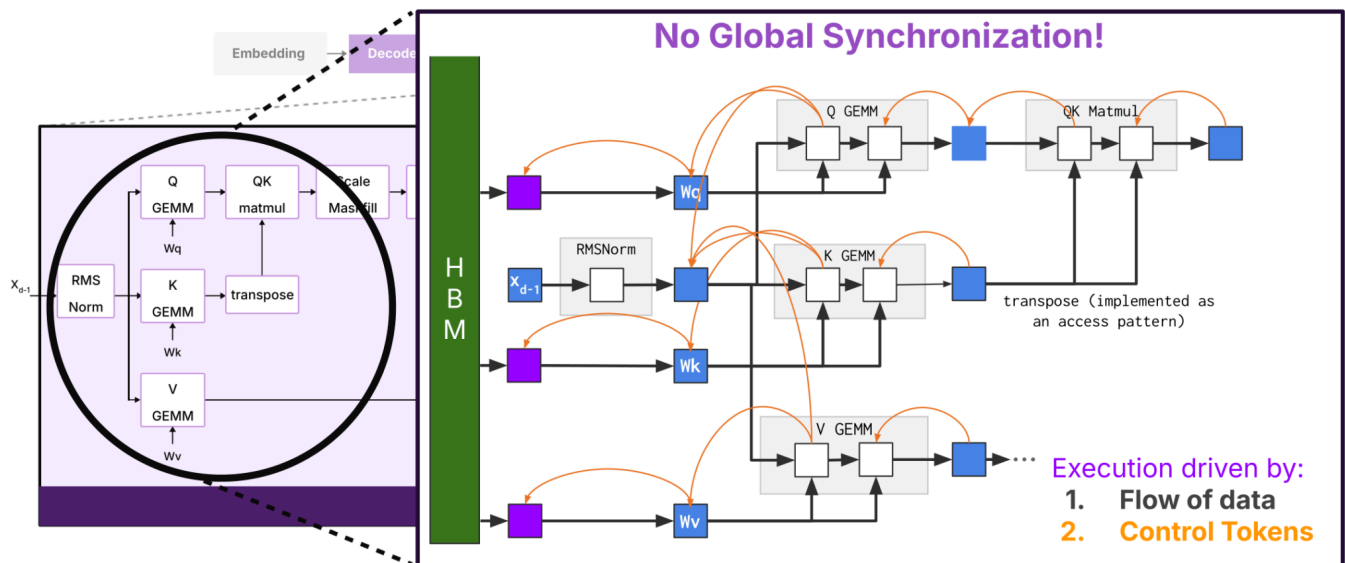
Compute - Communication Overlap: Within RDU



SambaNova SN50 Hot Chips 2026 Double-Buffered PMUs

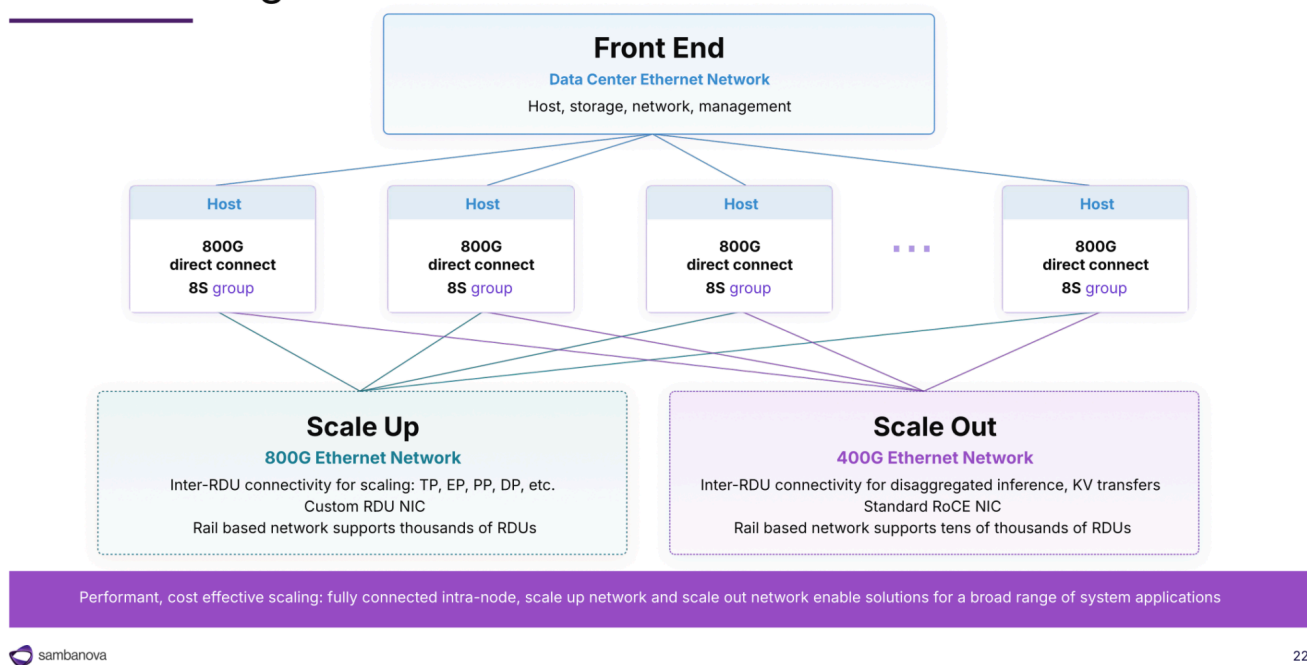
The SRAM amount used is not a function of the size of the model.

Compute - Communication Overlap: Within RDU



SambaNova SN50 Hot Chips 2026 No Global Synchronization

SN50 Scaling

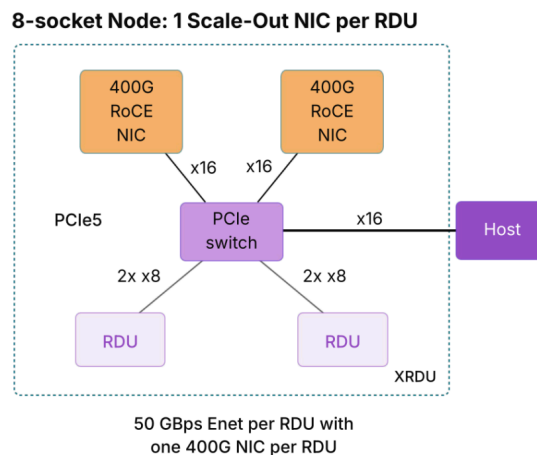
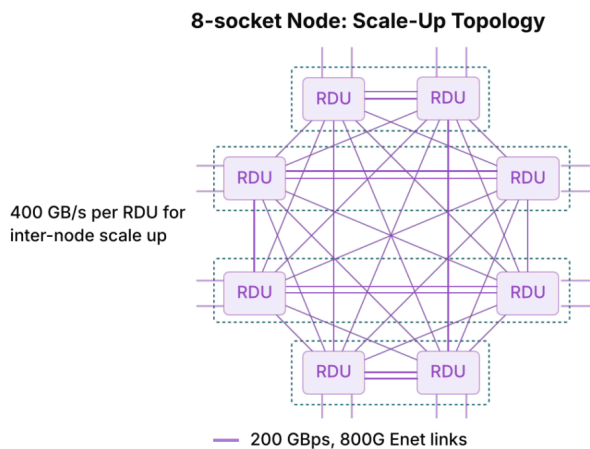


SambaNova SN50 Hot Chips 2026 SN50 Scaling

That was one RDU. How do things scale up for multiple RDUs? SambaNova employs both scale-up and scale-out networking. The Scale-up network is based on 800GbE, while scale-out is 400GbE. And then there is a front-end network.

SN50 Node: Scale-Up and Scale-Out

- RDUs have 10x integrated 800G Ethernet ports for scaling; **2 TB/s per RDU**
- 7 ports for fully connected 8S node scale up; 2 ports for scale up network
- 1x RoCEv2 NIC per SN50 RDU used for scale out between scale up domains

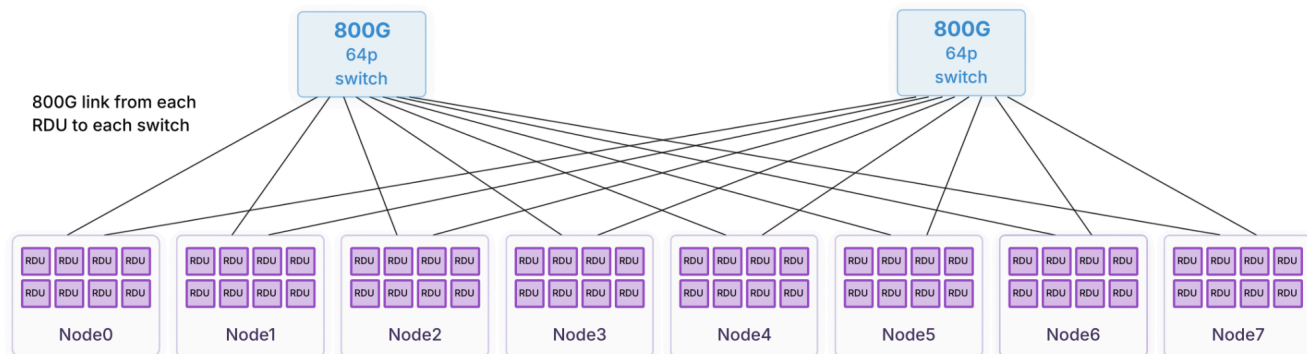


SambaNova SN50 Hot Chips 2026 Scale-up and Scale-Out

SambaNova uses an all-to-all topology for an 8 socket configuration.

Scaling Up Beyond One Node: 64-socket Example

- 64S SN50 scale up can be achieved with 2× 64p 800G switches
- Each of the 64 RDUs connects one 800G link to each switch



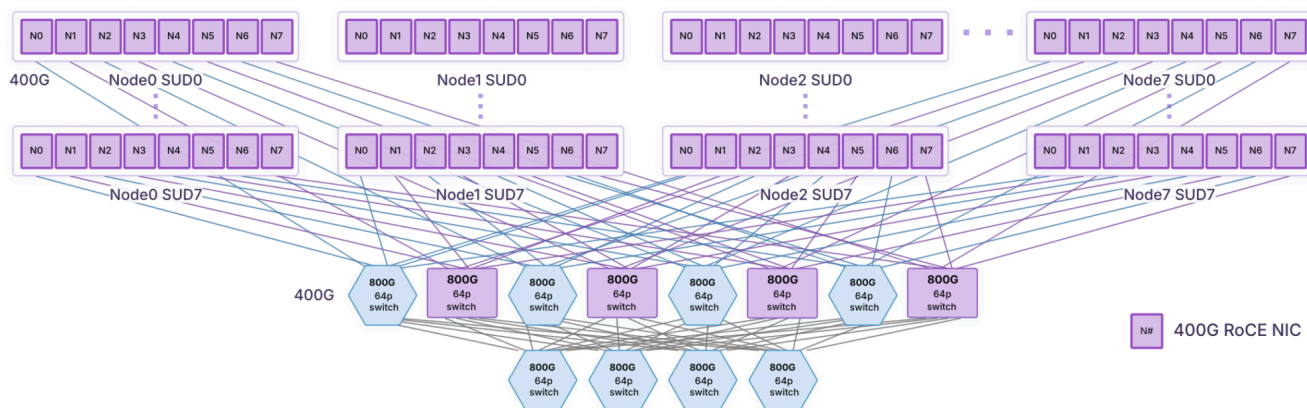
High bandwidth, single tier switching architecture for 64 sockets

SambaNova SN50 Hot Chips 2026 Scaling Beyond One Node

To go above 8 sockets, then the scale-up network is employed using Ethernet switches. Links are ganged, and every node is connected to each of two switches in this 64 chip socket configuration.

Scaling Out across Scale-Up Domains: 512 Socket Example

- Rail optimized 2-tier network
- 1× 400G RoCE NIC per SN50 RDU
- 512S scale out domain with 12× 800G 64p switches



Simple system scale out network

SambaNova SN50 Hot Chips 2026 512 Socket Example

Then things can be scaled out further, in this case employing both scale-up and out for a 512 socket configuration.

SN50 Supports All Forms Of Model Parallelism

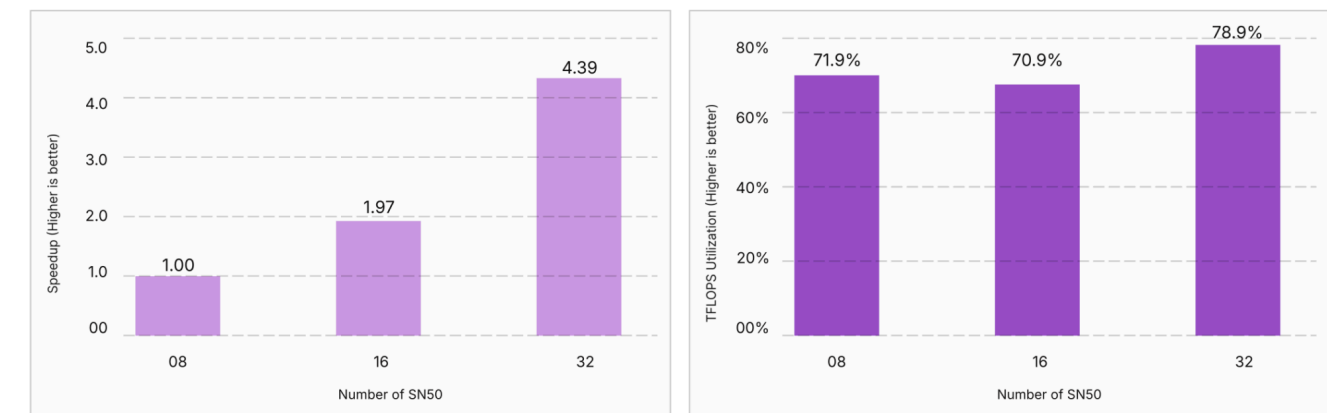
Model Parallelism	Communication Primitives			
Tensor Parallel (TP)	➡	Reduce-Scatter (RS)	+	All-Gather(AG) OR All-Reduce(AR)
Pipeline Parallel (PP)	➡	Send-Receive		
Expert Parallel (EP)	➡	All-to-All		
Data Parallel (DP)	➡	Reduce-Scatter (RS)	+	All-Gather(AG) OR All-Reduce(AR)

Tensor Parallel and Data Parallel resolve to the same collective pair
the same fabric path serves both, so scaling one does not require a second interconnect strategy.

SambaNova SN50 Hot Chips 2026 Model Parallelism

The key to performance on SN50 is overlap. SN50 supports all forms of model parallelism, and the collective communication forms that these models are built on.

TP: SN50 On-Silicon Tensor Parallel GEMM Benchmarking



Key takeaways:

- 70% or higher TFLOPs utilization consistently measured across 8, 16, and 32 sockets
- 32-SN50 achieves full overlap of compute and communication

SambaNova SN50 Hot Chips 2026 GEMM Benchmarking

Importance of Overlap - First Principles



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If you are able to overlap, you can do the compute and communications in parallel. That kind of overlap is not something GPUs can do.

TP: HBM and Compute Overlapped with Communication

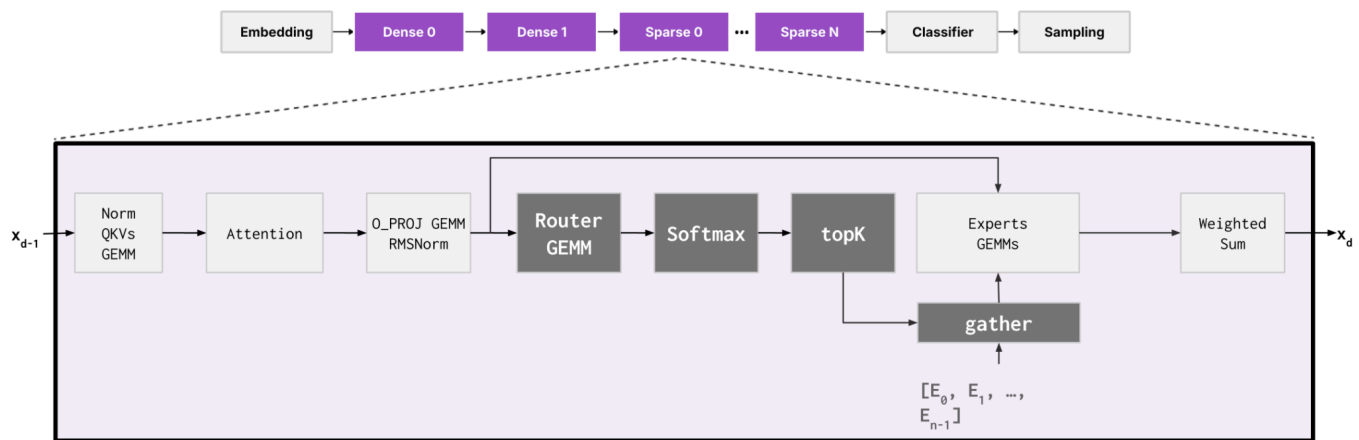


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SambaNova SN50 Hot Chips 2026 HBM and Compute Overlap

The building block for SambaNova is collective communication, which is the purple boxes in these diagrams. And the SRAMs can stream from one to another without having to go through a higher layer (e.g. HBM).

MoEs With TP

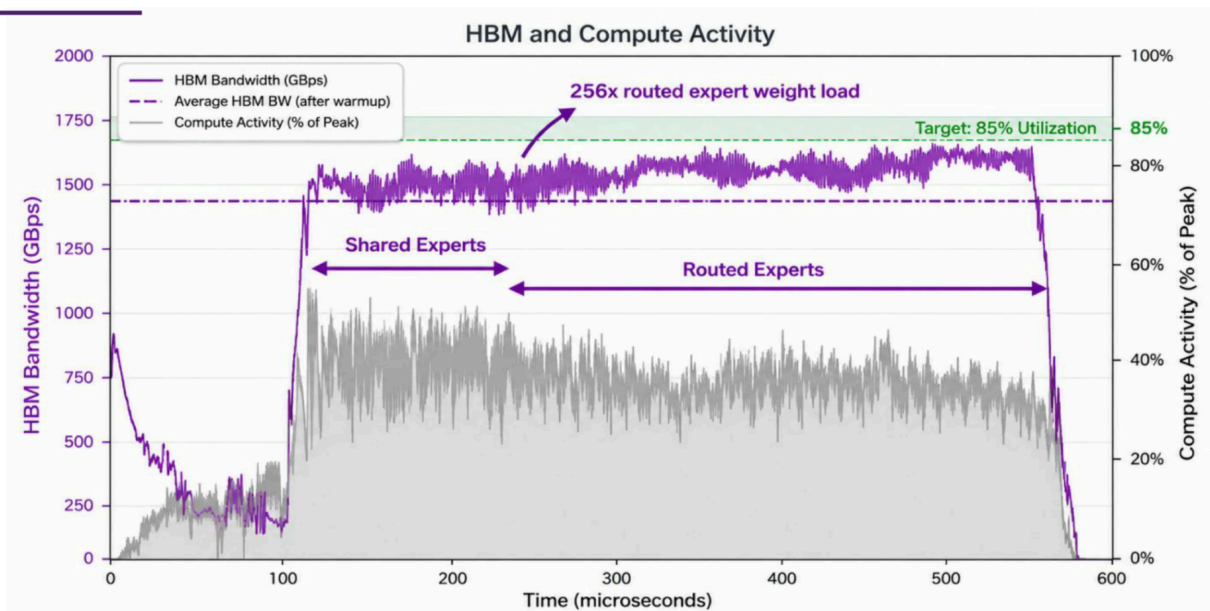


- MoEs are frequently mapped with TP-8, TP-16, and TP-32 on RDUs
- Kernel Looping extends to MoEs naturally with indexed weight loads
- Indexed loads work with both SRAM and off-chip memories (HBM, DRAM)
- To capture locality, experts of one layer are frequently prefetched into SRAM and indexed from there

SambaNova SN50 Hot Chips 2026 MoEs with TP

Here's a look at parallelism with tensor parallel.

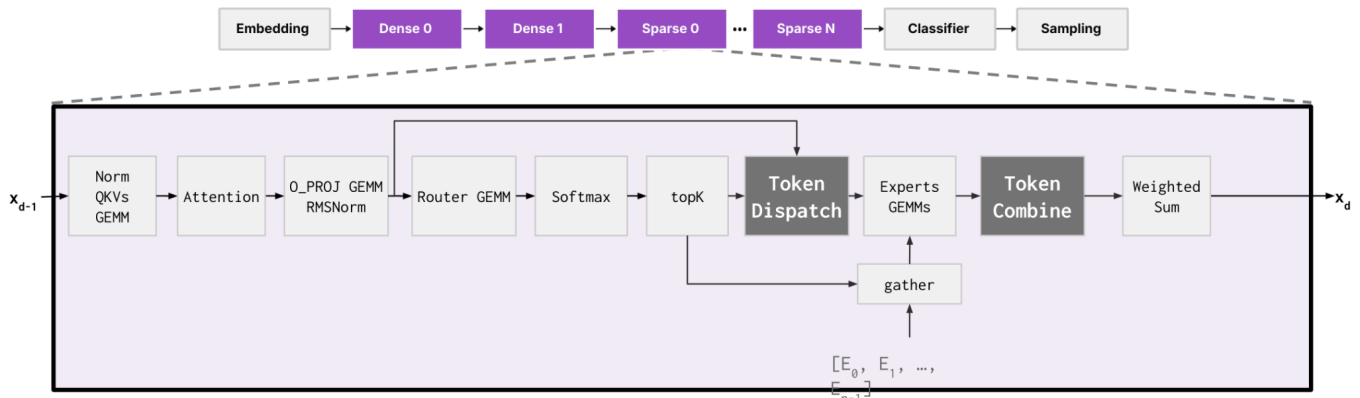
DeepSeek TP-32 MoE Layers Achieve >70% MBU



SambaNova SN50 Hot Chips 2026 DeepSeek TP-32

Here is a look at the bandwidth utilization that SN has achieved with DeepSeek.

MoEs With TP + EP

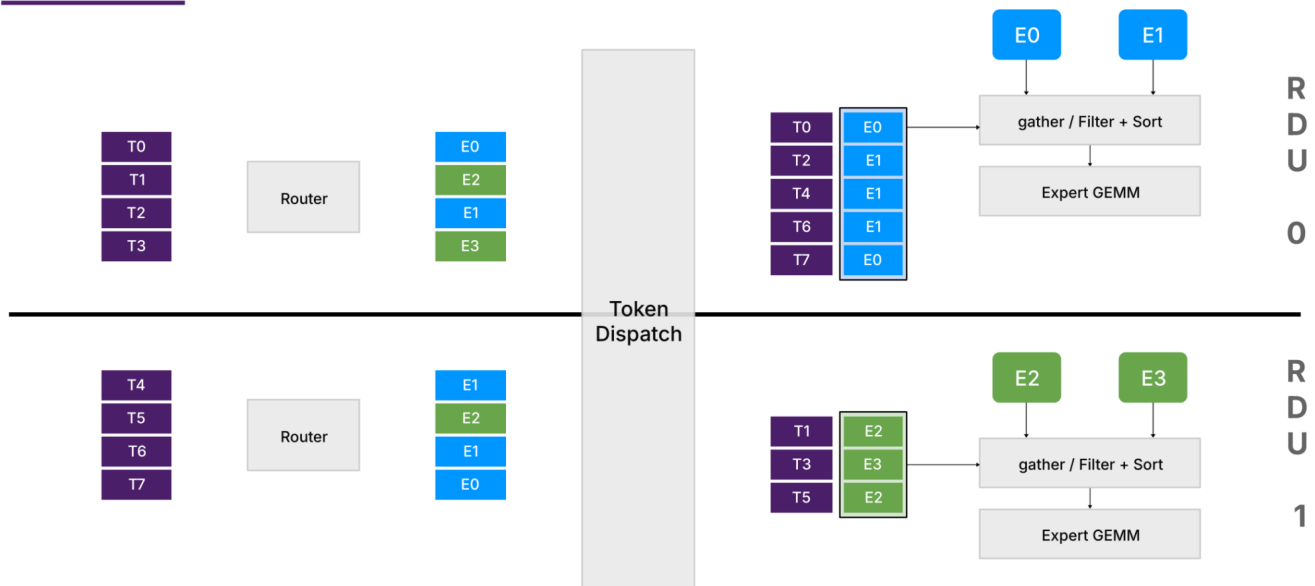


- EP introduces dynamic all-to-all communication across RDUs in **token dispatch** and **token combine**
- RDUs support two forms of token dispatch & combine: **broadcast, and all-to-all**
- Broadcast dispatch-combine: Uniform network traffic, extra network traffic, filtering at destination
- All-to-all dispatch-combine: Non-uniform network traffic, no extra network traffic, source filtering

SambaNova SN50 Hot Chips 2026 MoEs with TP and EP

Meanwhile they can also use expert parallel (EP) as an additional form of parallelism. This relies on broadcast-dispatch as well as all-to-all dispatch-combine.

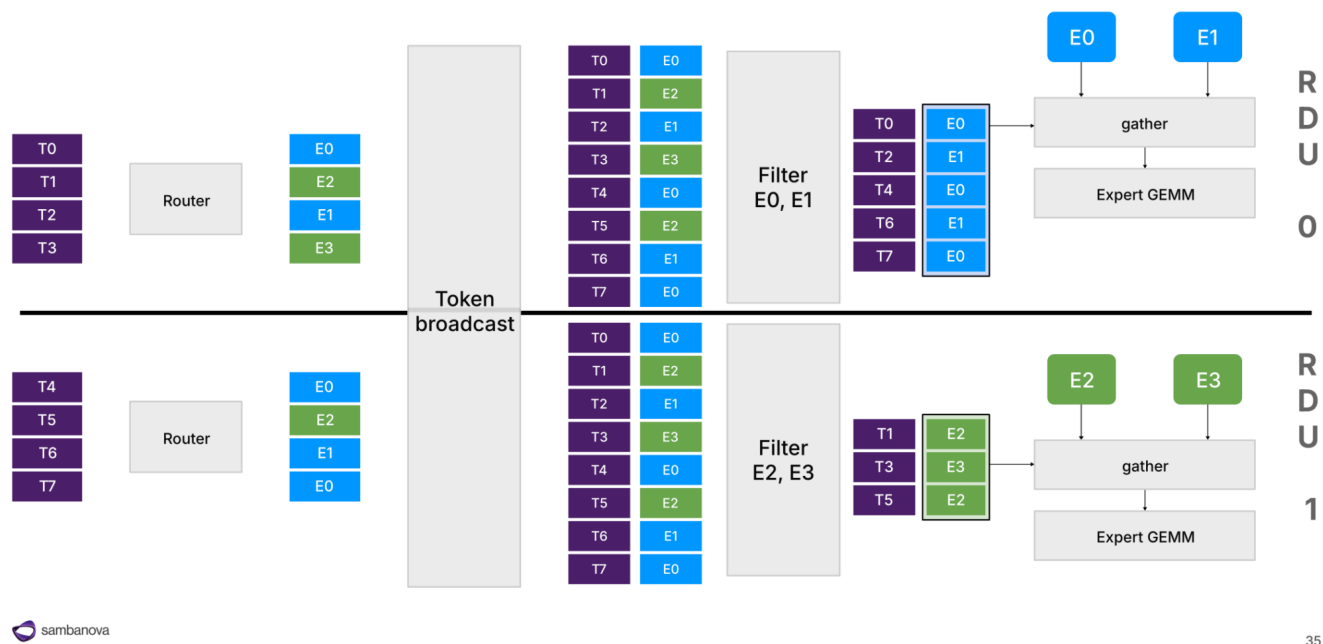
Token Dispatch: All-to-All



SambaNova SN50 Hot Chips 2026 Token Dispatch

With all-to-all, one way is to dynamically send everything to the target RDUs.

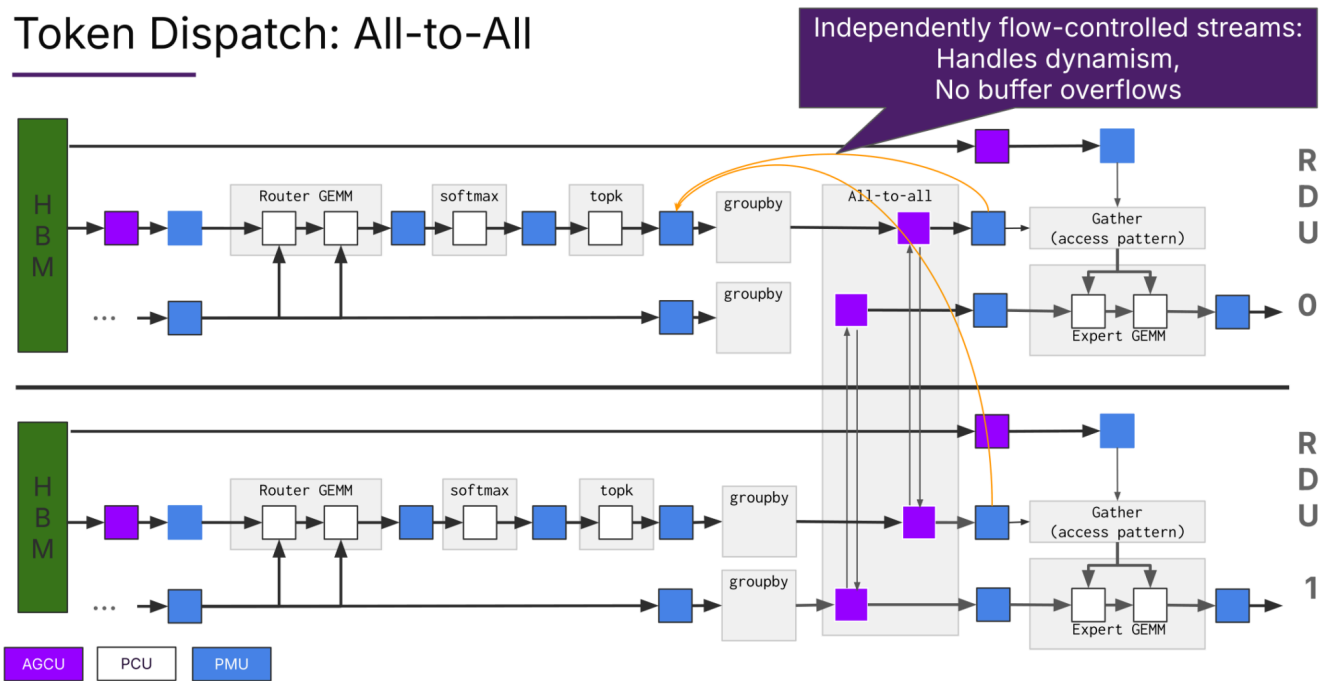
Token Dispatch: Broadcast + Filter



SambaNova SN50 Hot Chips 2026 Dispatch Broadcast

Alternatively, you can just blast everything to all of the RDUs and then filter out things afterwards.

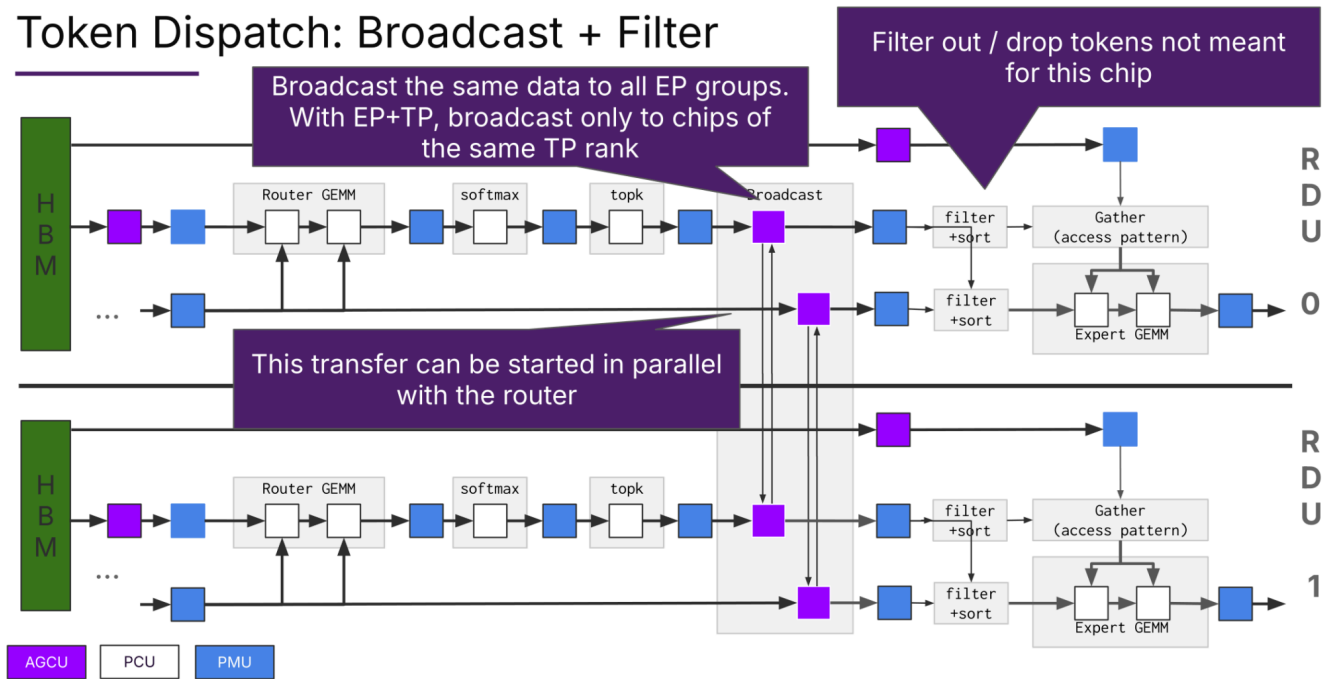
Token Dispatch: All-to-All



SambaNova SN50 Hot Chips 2026 Dispatch All-to-All

The all-to-all method requires a group-by operation at the end of the router to collect (group) the tokens before transmitting them SRAM-to-SRAM. All-to-all also means allowing dynamic traffic.

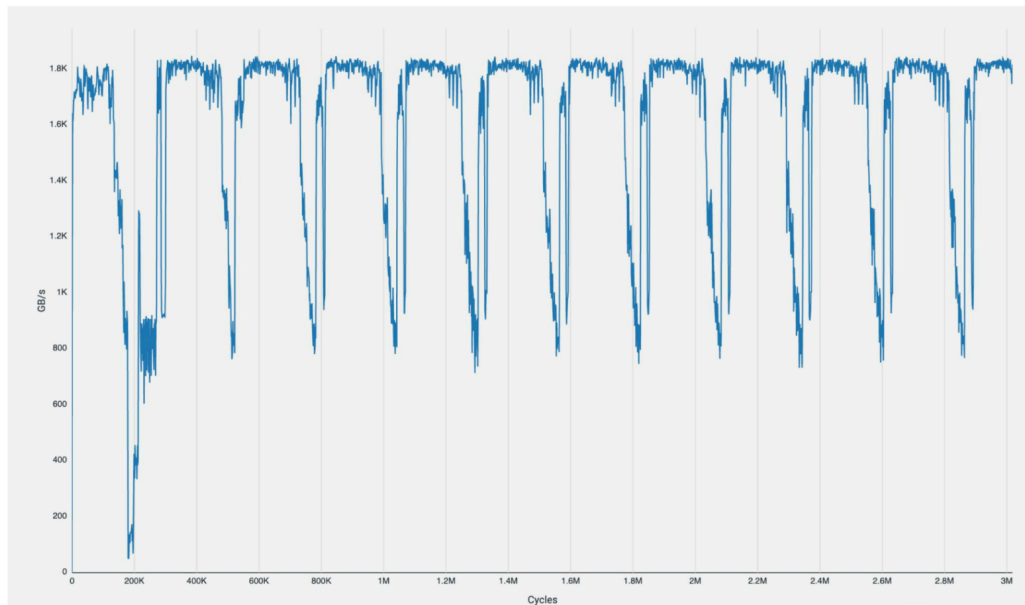
Token Dispatch: Broadcast + Filter



SambaNova SN50 Hot Chips 2026 Dispatch Broadcast and Filter

Now here's the other method of broadcast + filter. That is still an SRAM-to-SRAM operation, but with a filter operation on the PCUs of the receiving RDU. This keeps the network traffic parallel; though it does increase it a bit. And by not depending on the router, the transfer can be started early.

EP on 64-socket SN50: >70% Bandwidth To Load Experts

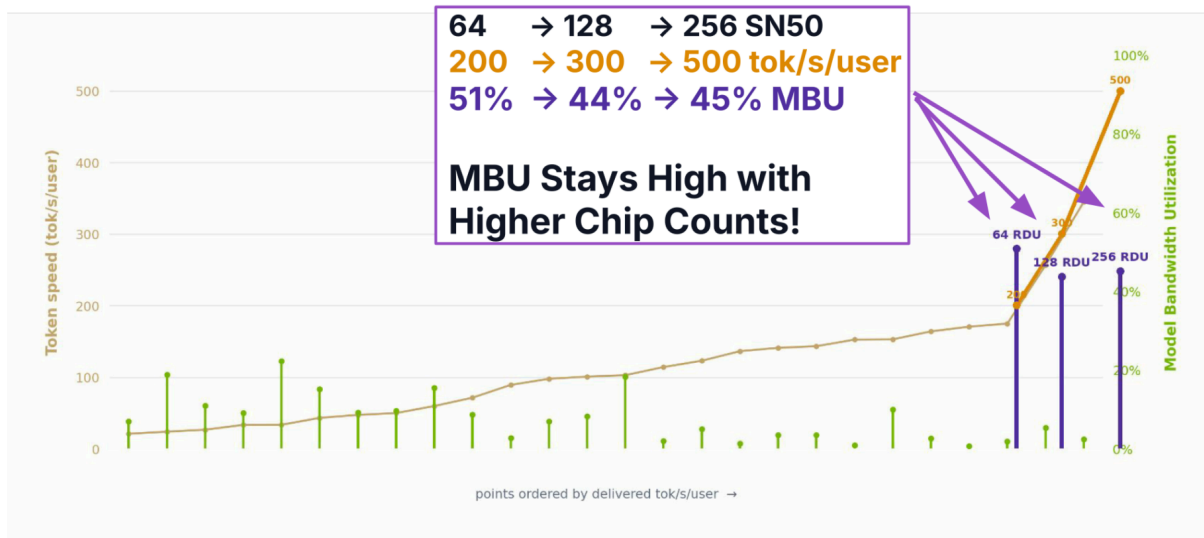


SambaNova SN50 Hot Chips 2026 EP on 64-Socket SN50

Here is another DeepSeek example, with SambaNova getting close to 80% bandwidth utilization for loading the experts in MoE.

SN50 RDU: High MBU At Scale

DeepSeek-R1 8K/1K | B300 FP4 | 26 InferenceX GPU points | sorted by delivered tok/s/user | RDU inserted by speed



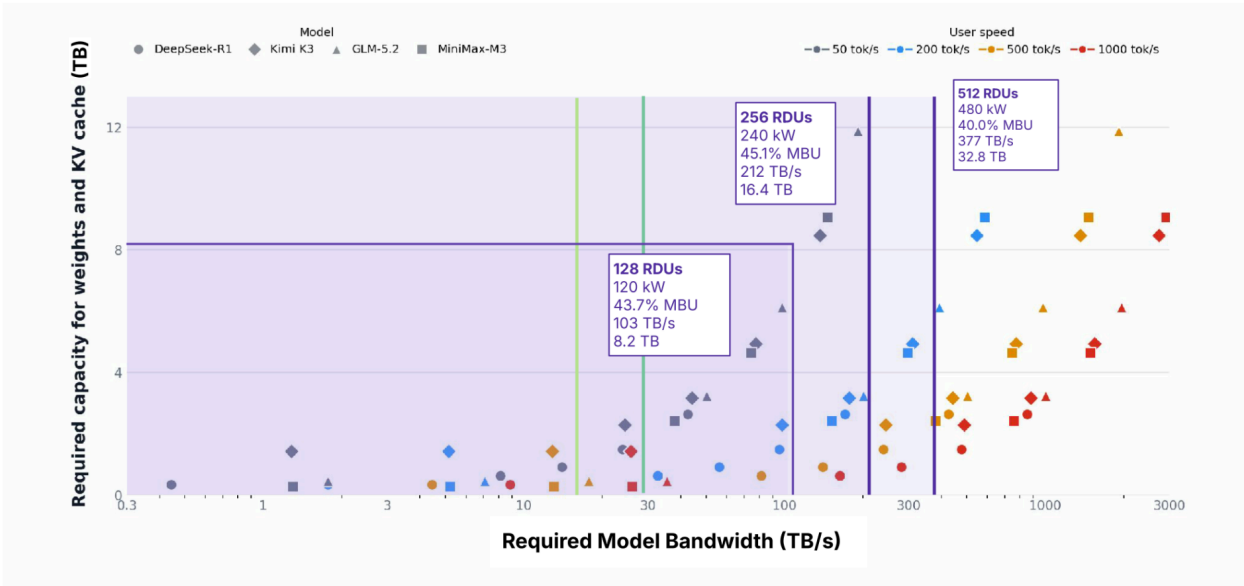
SambaNova SN50 Hot Chips 2026 High MBU at Scale

As a result of this, SN50 achieves a high MBU value even at scale, with MBU holding at 45% even with 256 SN50s. And this makes it possible to keep adding RDUs to scale up things even

further. This, in turn, means that models don't have to give up bandwidth.

SN50 RDU: Power Scales Capacity and Realizable Bandwidth

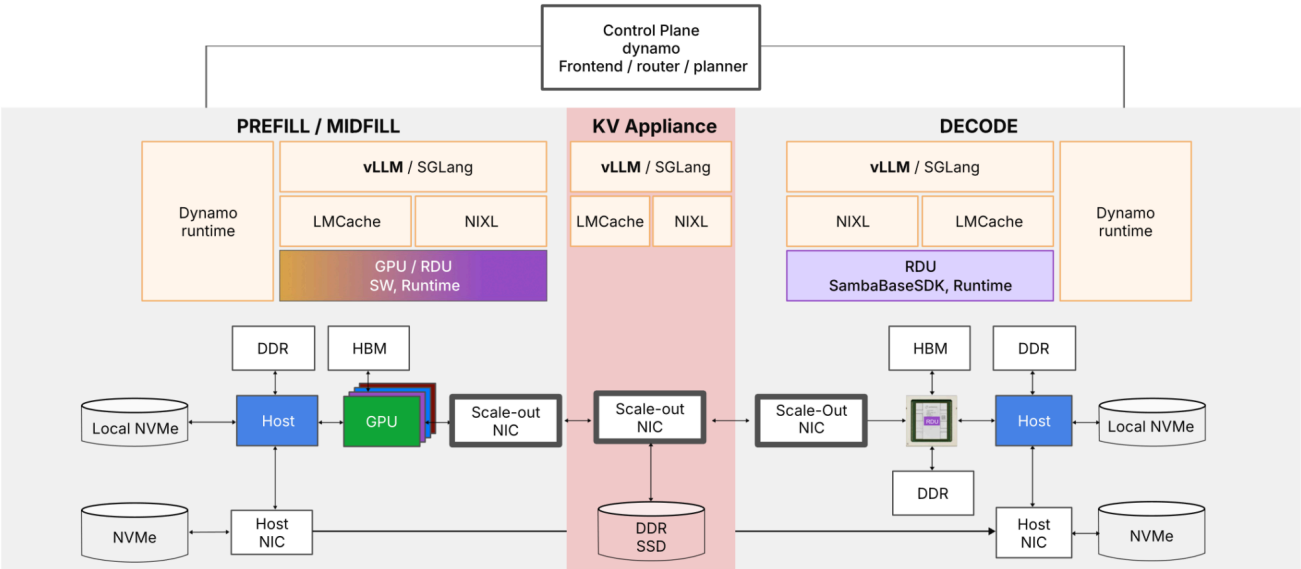
All frontier-model points | MTP3 accepted length = 3.1



SambaNova SN50 Hot Chips 2026 SN50 Power Scaling

Going back to SambaNova’s original chart about power scaling, here is what SN50 clusters of different sizes look like. A 512 RDU configuration is able to scale up to an aggregate model bandwidth capacity of over 350 TB/second. The systems can strongly scale, with MBUs still in the 40% range at 512 sockets.

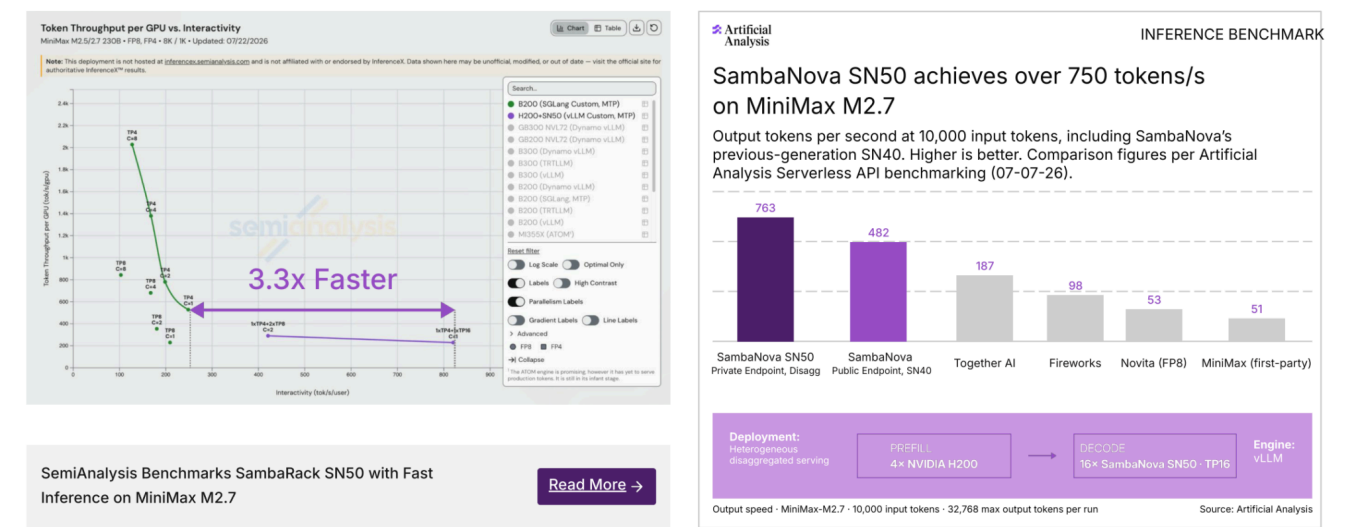
Putting it All Together: Heterogeneous Disaggregation



SambaNova SN50 Hot Chips 2026 SN50 Heterogeneous Disaggregation

Ultimately SambaNova is promoting a very similar picture as other dedicated inference chip firms, using one type of chips for prefill (and midfill), while using separate accelerators (i.e. SN50) for decode. Specifically, they’ve been using NVIDIA H200 + SN50, with RoCE for transferring between them.

Heterogeneous Disaggregation in Action



SambaNova SN50 Hot Chips 2026 SN50 Heterogeneous Disaggregation In Action

Finally, taking a look at that performance in action, based on an Artificial Analysis benchmark of SN50. The hardware achieves over 750 tokens-per-second in MiniMax M2.7.

